



AIME 2026 · OTTAWA

Where Evidence Meets the Patient

Knowledge graphs, clinical reasoning, and the messy gap to real patient data

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Three things people mean by “knowledge graph” in health care

It's a broad term. This talk is about the last kind and the gap to the other kinds is the whole story.

ALSO EXISTS

Patient data

One specific patient's record as a graph — the “spiderweb” around the person.

patient-centric KGs (PCKGs)

ALSO EXISTS

Terminology

The vocabularies themselves — code systems and value sets as graphs.

SNOMED hierarchy · interface terms

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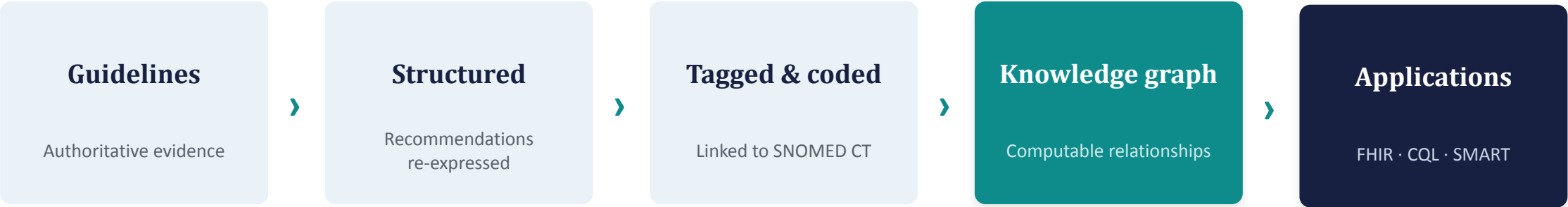
Knowledge

Encoded clinical knowledge — general, reusable, evidence-linked.

BMJ's graph · SNOMED CT · UMLS

From evidence to computable graph

An editorial pipeline — each step done by the specialists best placed to do it.



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Acute exacerbation of chronic obstructive pulmonary disease [Ver contenido en español](#)

OVERVIEW	THEORY	DIAGNOSIS	MANAGEMENT	FOLLOW UP	RESOURCES
Summary	Epidemiology Aetiology Case history	Recommendations History and exam Investigations Differentials Criteria	Recommendations Treatment algorithm Prevention Patient discussions	Monitoring Complications Prognosis	Guidelines Images and videos References Patient information Calculators Evidence

Treatment algorithm

Treatment algorithm

ACUTE

on presentation

1ST LINE

short-acting bronchodilator →

CONSIDER

systemic corticosteroid →

ACUTE

on presentation

1ST LINE

short-acting beta-2 agonist →

CONSIDER

oxygen →

CONSIDER

ventilation →

CONSIDER

antibiotic therapy →

CONSIDER

supplemental treatment →

ACUTE | on presentation

1ST LINE

short-acting bronchodilator

Give a short-acting beta-2 agonist (e.g., salbutamol) at an increased dose or frequency from the patient's usual baseline treatment as first-line therapy.[1][90]

For a moderate to severe exacerbation, use a nebuliser driven on air (not oxygen). [90]

This is to avoid worsening hypercapnia.[90]

The driving gas for nebulised therapy should always be specified in the prescription.[90]

For a mild exacerbation, use a metered-dose inhaler (MDI) plus spacer.

People with severe dyspnoea with low inspiratory flow rates may have difficulty achieving proper technique and medication delivery from the MDI; a nebuliser may be easier for these patients to use.

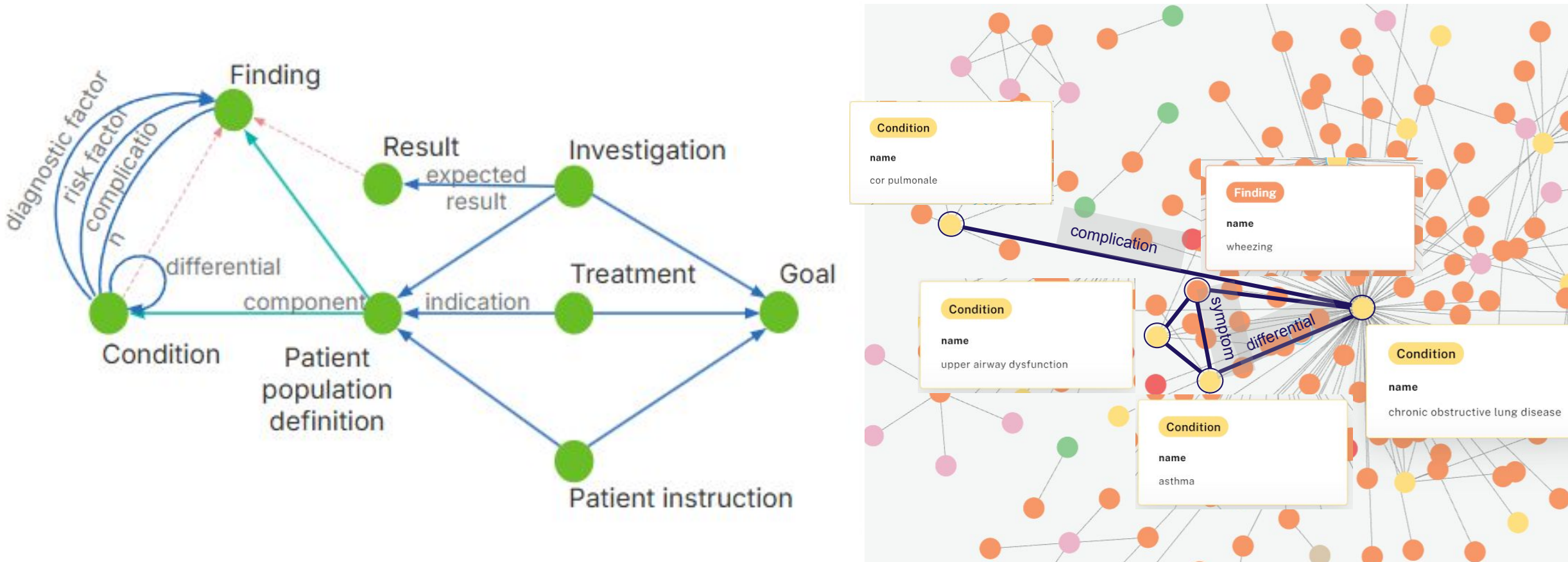
You may see benefit from this initial treatment within 30 minutes.

See Coronavirus disease 2019 (COVID-19).

3 of 18

One condition, a whole clinical model

BMJ's clinical knowledge graph schema — every finding typed and linked to its evidence.



Left: the schema — Finding, Condition, Result, Investigation, Treatment, Goal, joined by typed edges (diagnostic factor, risk factor, complication, differential). Right: the same model instantiated as one condition's real, dense graph.

Reasoning is traversal, not generation

The graph doesn't guess an answer. It walks coded relationships until a pattern is complete.

THE GRAPH

- 1 Match coded findings to typed nodes
- 2 Combine them against the criteria on each edge
- 3 When a pattern completes, assert the conclusion
- 4 Carry the provenance — which evidence, which guideline

AN LLM ALONE

Predicts a fluent, plausible answer from patterns in its training.

- No traceable path to evidence
- Can't show which criteria were met
- Can't be audited or constrained
- Right until it is confidently wrong

Determinism is the point. *The same record gives the same answer every time, and the answer comes with its working attached.*



ACT I • WHY ONE GRAPH DOES MANY THINGS

Same machinery, different question

Diagnosis, investigation and treatment aren't separate features. Each is the same traversal over the same graph — asking a different typed edge.

Diagnose

do the findings complete a pattern?

edge: diagnosticFactor

Recommend tests

what evidence is still needed?

edge: inv

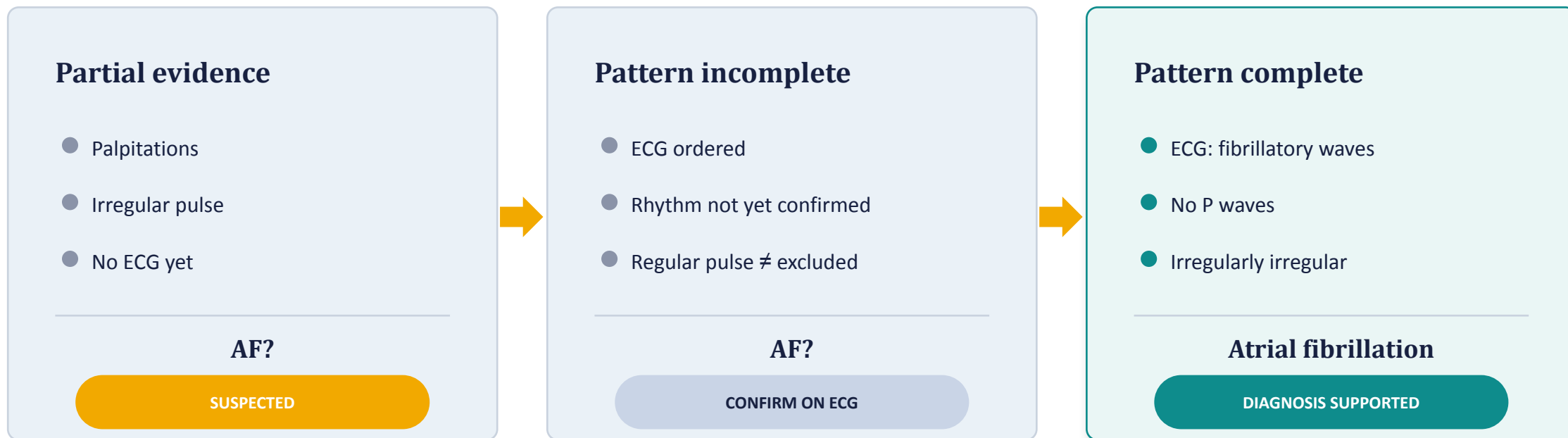
Treat safely

what does this patient's context allow?

edge: action

Diagnosis: suspicion is not diagnosis

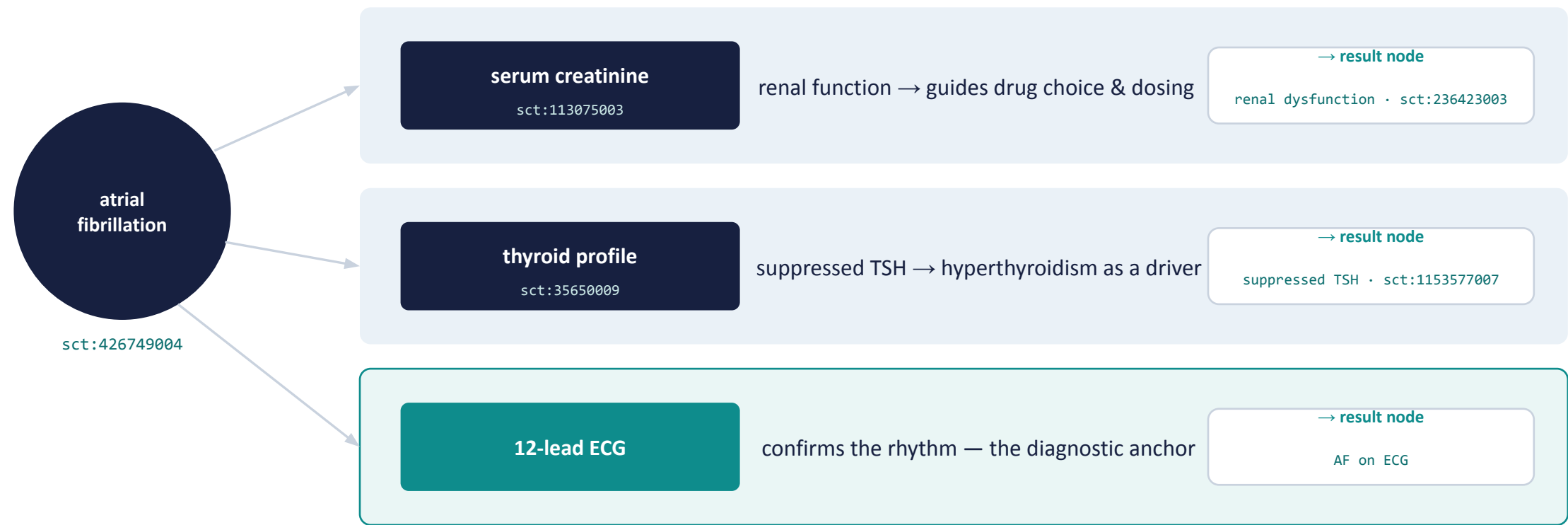
The graph can hold a candidate and name exactly what evidence is still missing.



The graph withholds a diagnosis it can't support — and tells you the one test that would resolve it. That is reasoning you can defend.

Investigation: what to check, and why

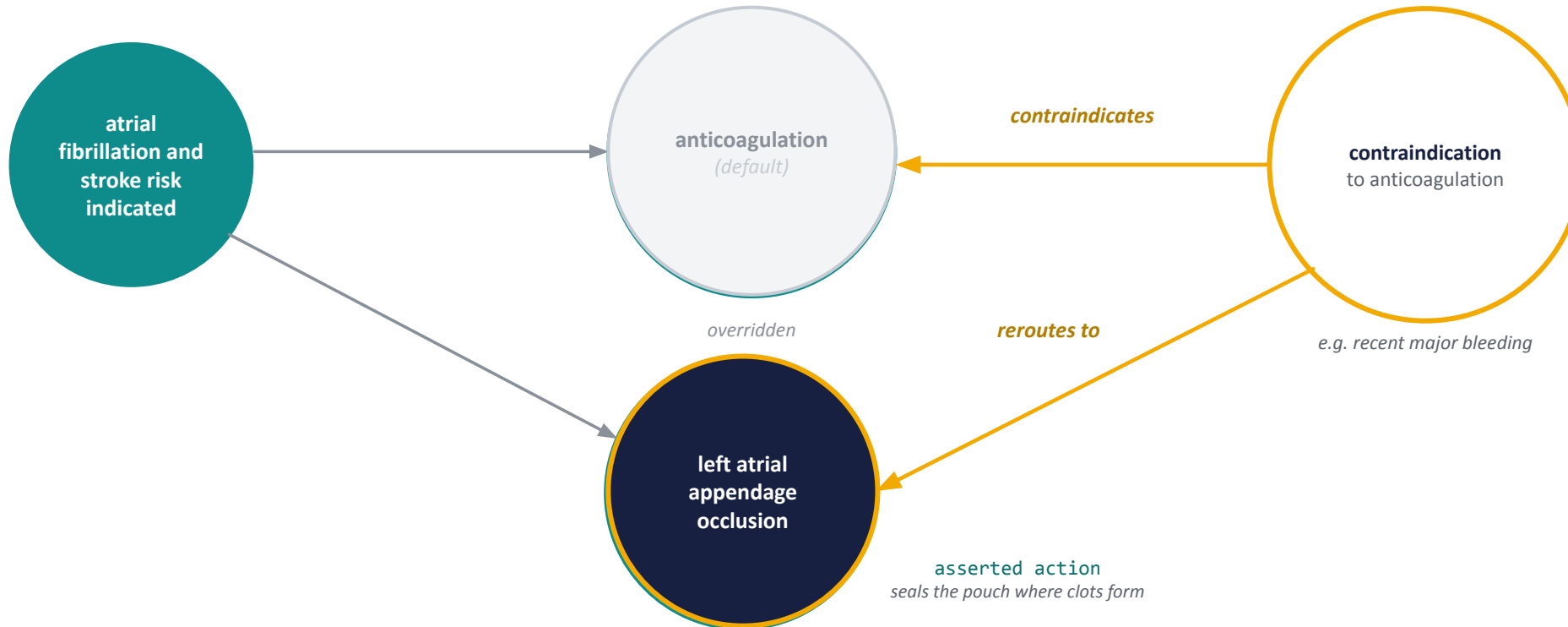
Traverse from the condition to the investigations the guideline ties to it — each one coded.



Not a static checklist. The investigations are tied to this condition's findings, and each result feeds back as a node the reasoner can use.

Treatment: the right action, routed

Same traversal, over treatment edges: a contraindication edge overrides the default and reroutes to a different action.



Deterministic routing, traceable to the 2023 ACC/AHA, 2020 ESC and NICE AF guidelines the node cites — the WATCHMAN™ device is non-inferior to warfarin (PROTECT AF).



ACT II • THE HARD PART

Everything so far assumed clean data

In a vignette, the findings arrive already coded and tidy. In a real EHR or patient notes they don't. Our early work on real records keeps surfacing the same thing:

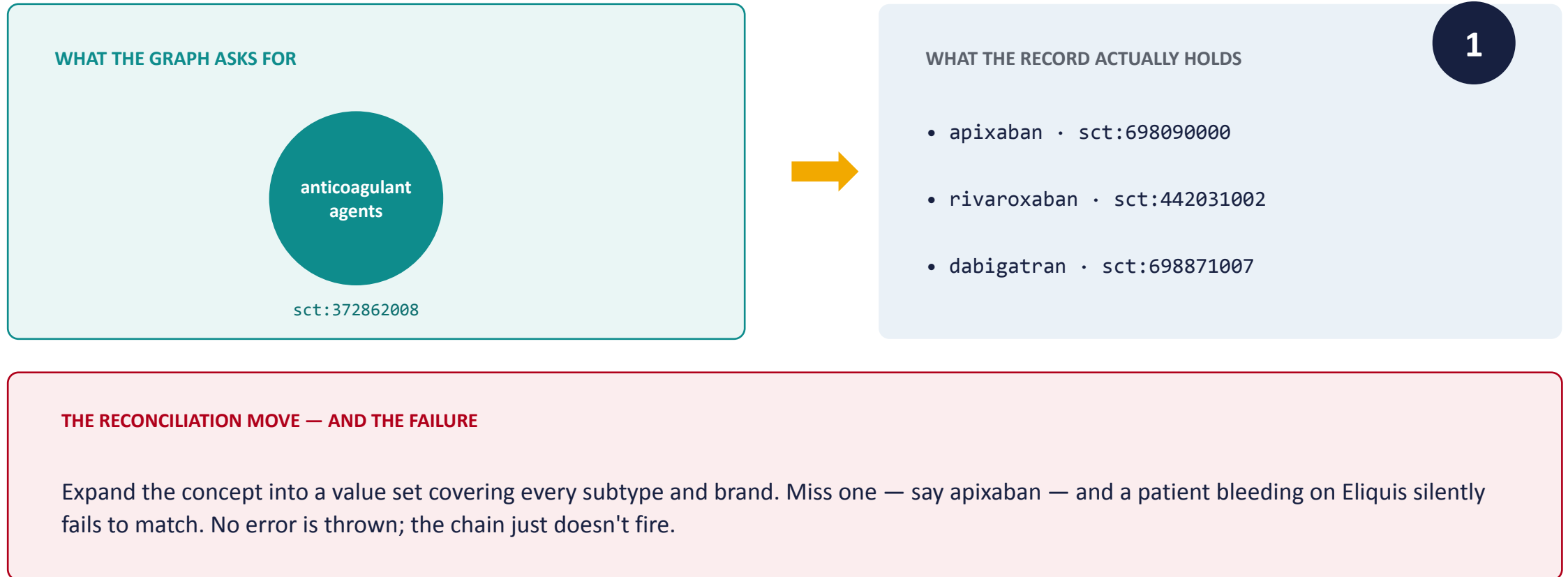
There are 4 gaps in matching patient records to a clinical reasoning KG

Early work in our group on EHR integration

Each must be closed before reasoning can run.

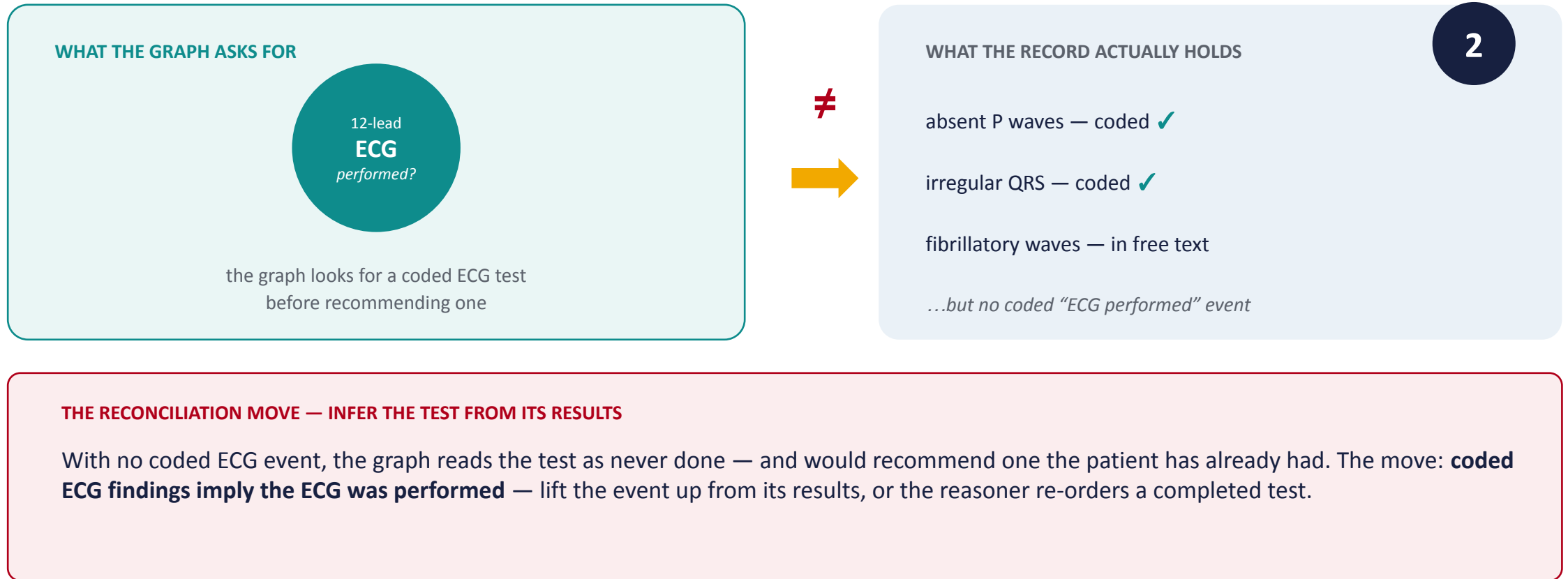
Vocabulary: same thing, different code

The graph stores one high-level concept; the record names the specific drug.



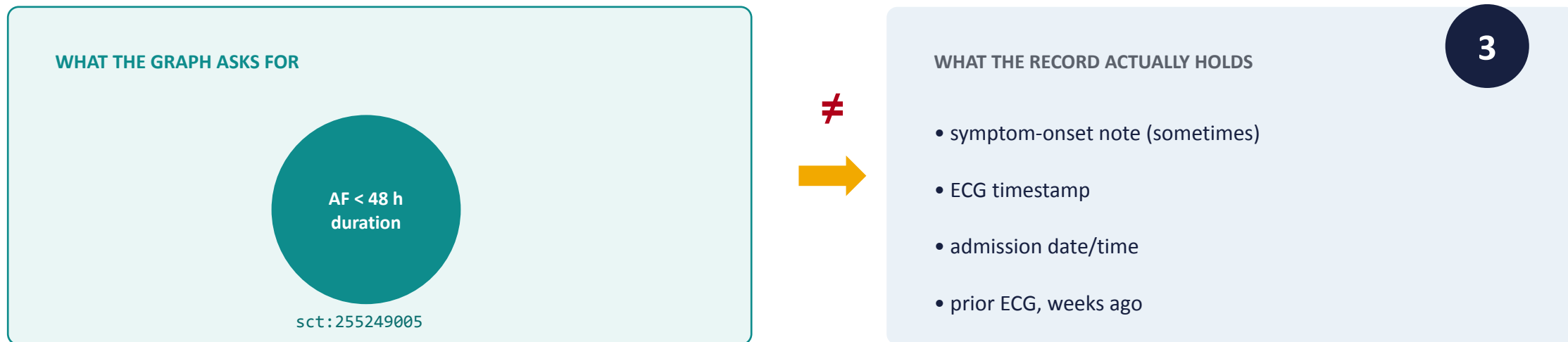
Abstraction: a result without its test

Recommending investigations, the graph checks whether an ECG was done — the record codes the ECG's results, not the test itself.



Temporal: one node, scattered evidence

The graph encodes a single computable duration. The record never stores it directly.

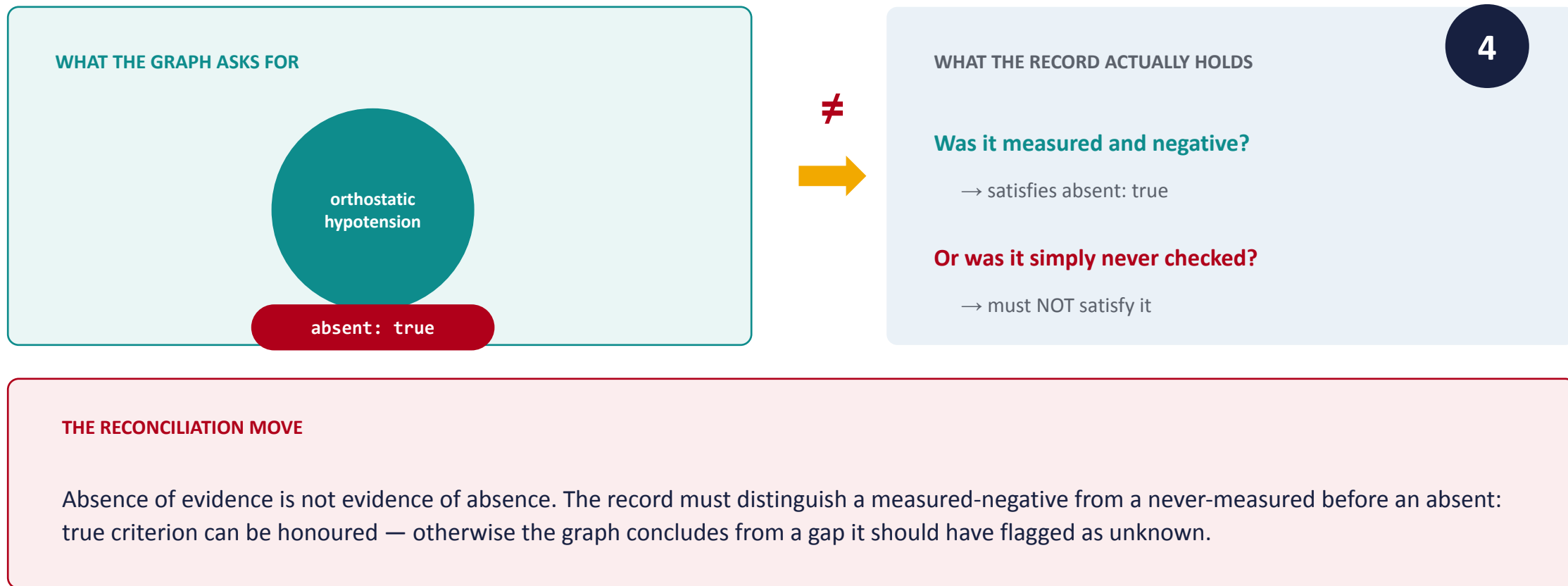


THE RECONCILIATION MOVE — AND WHY IT BITES

The criterion is one node; the evidence is a temporal pattern that must be computed by differencing timestamps across the record. Get it wrong and the patient is cardioverted unsafely, or anticoagulated unnecessarily — the < 48 h boundary is a real clinical decision point.

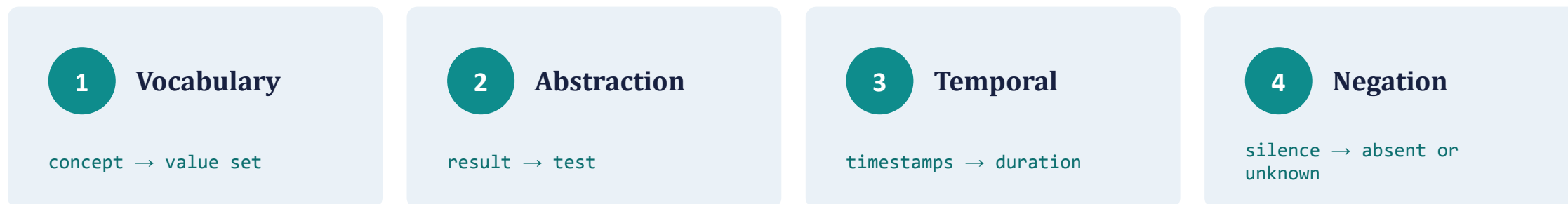
Negation: absent is not the same as unknown

The graph asserts a finding must be absent. The record's silence is ambiguous.



One move, made four times

Every gap is the same act: make the graph's implicit knowledge explicit against messy data — before reasoning.



coded patient data → **RECONCILIATION** → graph-shaped findings → reasoning

The graph holds knowledge declaratively — nested, typed, conditional. Reconciliation is the step that unfolds it against a specific patient. Skip it, and deterministic reasoning has nothing clean to run on.

Where should reconciliation live?

We don't think this is solved. The design choice itself is the research question.

OPTION A · A PRE-PROCESSING LAYER

A layer rewrites the record into graph-shaped findings before the reasoner runs.

- + Reasoner stays clean and simple
- + One place to test and audit
- A second model of the world to maintain

OPTION B · SMART NODES

Each node carries its own logic for how to satisfy itself from a record.

- + Logic lives with the knowledge it serves
- + Updates with the guideline, in one place
- The graph gets heavier and harder to reason over

Our current work — value sets, and agents that map vignette concepts onto them — sits mostly on the **pre-processing side**. The EHR work is now testing **whether that holds for real records**.



WHERE THIS LEAVES US

Reasoning is the easy half

A clinical knowledge graph reasons deterministically, traceably, auditably — once the data is in its shape. The underestimated problem is getting it there.

The graph

deterministic, provenance-carrying
reasoning

The gap

four structural mismatches with real records

The work

reconciliation — where graph meets patient



Thank you

Where evidence meets the patient: knowledge graphs, reasoning, and the reconciliation gap

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Questions & discussion welcome
& please keep in touch!

